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**MCS-011**

**M. C. A. (REVISED)/B. C. A. (REVISED)**  
**(MCA/BCA)**

**Term-End Examination**

**June, 2020**

**MCS-011 : PROBLEM SOLVING AND  
PROGRAMMING**

*Time : 3 Hours*

*Maximum Marks : 100*

*Weightage : 75%*

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**Note :** (i) *Question No. 1 is compulsory.*

(ii) *Answer any three questions from the rest.*

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1. (a) Give the equivalent C expression for the following algebraic expression :  $2 \times 3 = 6$

(i) 
$$\frac{a^2b^3c^4 - d^4}{e(m - n)}$$

(ii) 
$$xy - \left[ (p + q)^4 / r^2 \right]$$

**P. T. O.**

- (b) Write a recursive function in C to generate a Fibonacci series. 6
- (c) Write an algorithm and draw a corresponding flow chart to check whether the given number is prime or not. 8
- (d) Write an interactive C program to perform the following operation on a  $3 \times 3$  matrix with appropriate validation checks : 10

$$C = A * B$$

- (e) Explain the differences between static, auto, register and global variables with an example for each.  $2\frac{1}{2} \times 4 = 10$

2. (a) List the arithmetic, logical and relational operators in C. When  $a$ ,  $b$ ,  $c$  and  $d$  are integers and values of  $a$ ,  $b$  and  $c$  are 8, 6 and 4 respectively. Find the value of  $d$ , 8
- if  $d = a + (b - c) * a / c$

- (b) Write a program in C to generate the following pattern : 6

1

1 2

1 2 3

1 2 3 4

- (c) Differentiate between call-by value and call-by reference using an example program for each. 6

3. (a) Using pointers, write a program in C to count the no. of occurrence of each character in a given string. 10

Sample I/P : ARRANGE

O/P :

A - 2 times      G - 1 time

R - 2 times      E - 1 time

N - 1 time

- (b) Write a C program to calculate the income tax for 5 employees if name of the employee, designation, department and annual salary is given as per the following slab-rate : 10

Below 5 lakhs : No tax

Above 5 lakhs and : 10% tax on  
below 7 lakhs annual income

Above 7 lakhs and : 20% tax on  
below 10 lakhs annual income

Above 10 lakhs : 30% tax on  
annual income

4. (a) Using file handling concept, write a C program to read a file and count the no. of lines in the file. 10

- (b) What are Unions ? Give an example code segment to initialize a union and to access a member of a union. Mention the difference between a structure and a union.

10

5. (a) Explain the following conditional statements with an example for each : 10

(i) If ..... else statement

(ii) If statement

(iii) Nested if ..... else statement

(iv) Else ..... if ladder

- (b) Define a function in C. List and explain various categories of functions. Also, illustrate a function to find square root of a given number.

10